44 Osmoregulation and Excretion

KEY CONCEPTS

Figure 44.1 At 3.5 m, the wingspan of a wandering albatross (*Diomedea exulans*) is the largest of any living bird. This massive bird remains at sea day and night throughout the year, returning to land only to reproduce. Despite drinking only seawater, the albatross keeps its body salts and water in balance.

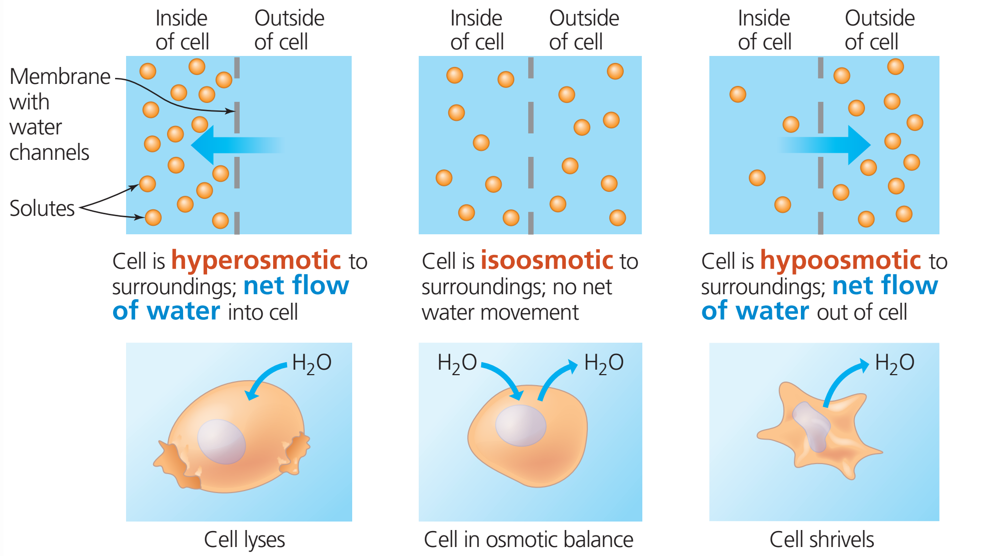
Study Tip

Make a table: As you read Concept 44.2, complete a table like the one shown to help organize information about the major forms of nitrogenous waste. In the first three rows, use the terms "high," "medium," or "low" to describe the relevant properties of these forms.

Waste attribute	Waste product		
	Ammonia	Urea	Uric acid
Toxicity			
Energy cost to produce			
Water loss during excretion			
Example of organism excreting this product			

How do animals regulate water and salt concentrations in their tissues?

Lacking a means to pump water across cell membranes, animals transport salts and thereby direct water movement into or out of cells by **osmosis**, in which water undergoes net diffusion from an area of higher free H_2O concentration (lower solute concentration) to an area of lower free H_2O concentration (higher solute concentration).



As you'll read, excretory systems not only maintain salt and water balance, but also rid the body of **nitrogen-containing wastes**.

Go to Mastering Biology

For Students (in eText and Study Area)
- Get Ready for Chapter 44
- Figure 44.14 Walkthrough: How the Human Kidney Concentrates Urine
- BioFlix® Animation: Membrane Transport

For Instructors to Assign (in Item Library)
- Tutorial: Kidney Structure and Function
- Scientific Skills Exercise: Describing and Interpreting Quantitative Data